



Medical Microbiology

Third stage- College of Dentistry

Lab. 15

Virology

Prepared by:

PhD. Dheyaa Naji Hamza

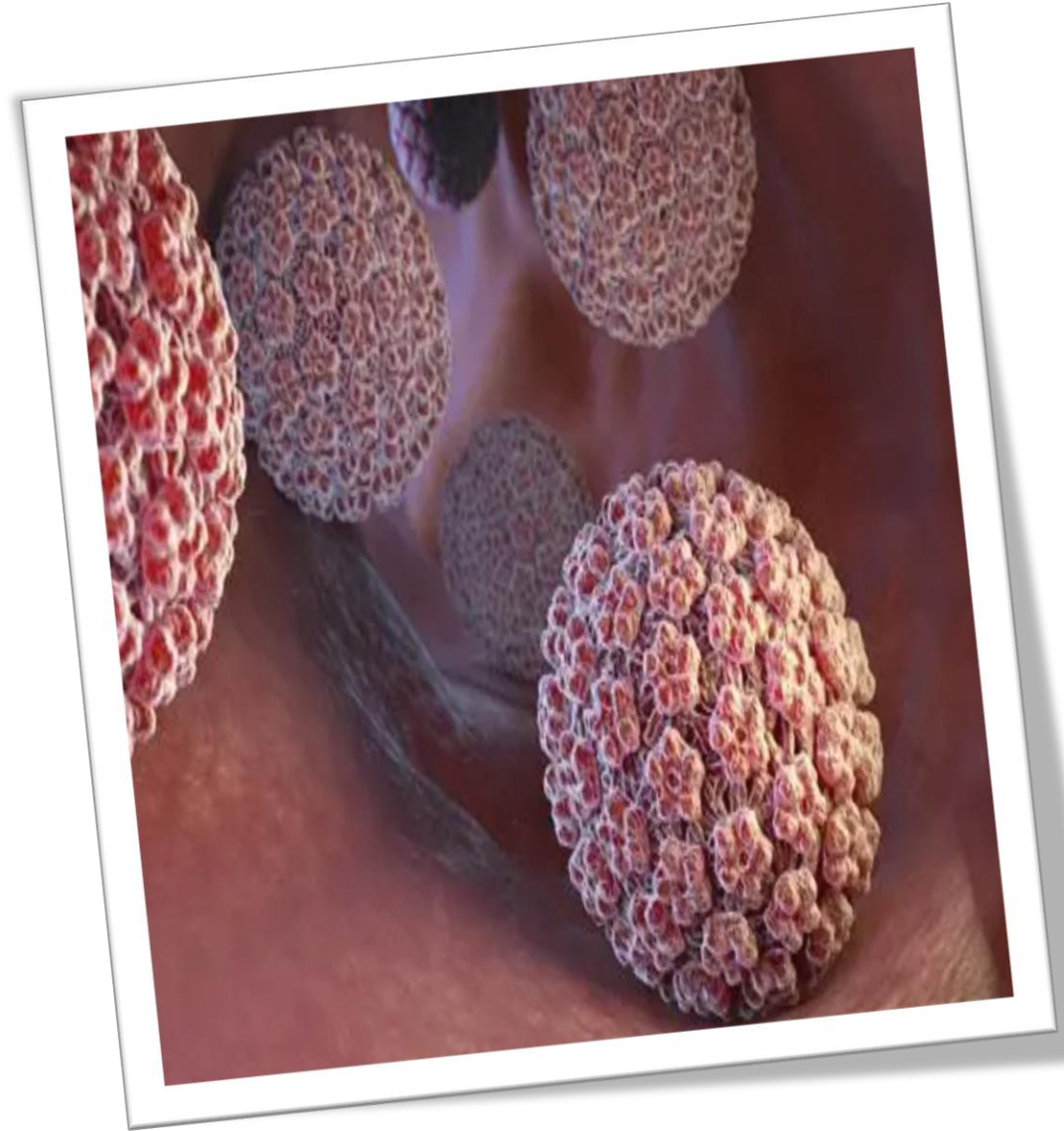
M.Sc. Zainab Majid Mohammed

Learning Objectives:

After completing this lab, you should be able to:

1-Describe the microscopic morphology and general Characteristics of the (HSV) and (HIV).

2-Explain the laboratory techniques used for the detection and identification of (HSV) and (HIV).



Introduction:

- Viruses are extremely small infectious agents that can only replicate inside living host cells.
- They are considered **obligate intracellular parasites** because they depend completely on the host cell machinery for reproduction.
- Viruses are much smaller than bacteria and cannot grow on artificial culture media.
- A virus is mainly composed of **genetic material**, which may be **DNA or RNA**, surrounded by a protective protein coat called the **capsid**.
- **Some** viruses also possess an outer **envelope** derived from the host cell membrane.
- Viruses infect different types of organisms including humans, animals, plants, and bacteria.

Herpes Simplex Virus (HSV):

Classification and Morphology

Genetic Material: Linear double-stranded DNA (dsDNA).

Capsid: Icosahedral protein shell consisting of 162 capsomeres.

Tegument: An amorphous protein layer between the capsid and the envelope.

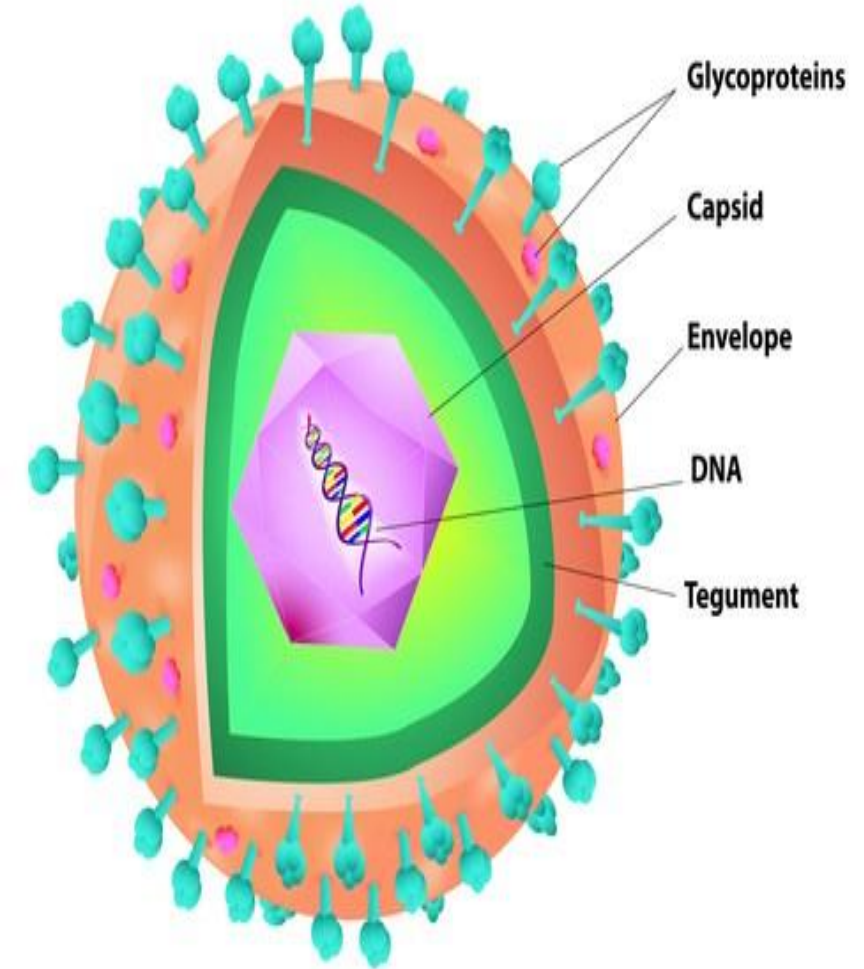
Envelope: A lipid bilayer derived from the host cell membrane, containing glycoproteins (essential for viral attachment and entry).

Main Types:

HSV-1: Primarily associated with **orofacial** infections.

HSV-2: Primarily associated with **genital** infections.

Structure of the Herpesvirus virion



Clinical Manifestations in Dentistry:

Primary Herpetic Gingivostomatitis: Usually occurs in children and causes painful blisters in the mouth, often accompanied by fever and swollen lymph nodes.

Recurrent Herpes Labialis: Repeated episodes of “cold sores” that appear on the border of the lips.

Laboratory Diagnosis:

1-Tzanck Smear: Detects multinucleated giant cells from vesicle scrapings.

2-Viral Culture: Virus grown in cell culture producing cytopathic effects within 48–72 h.

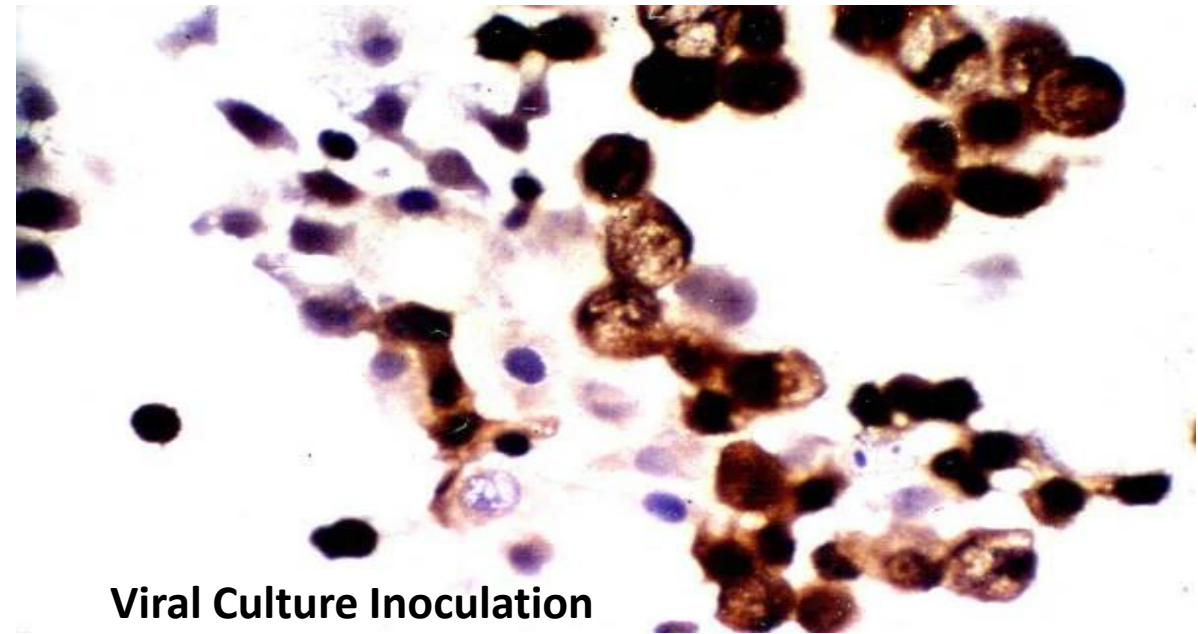
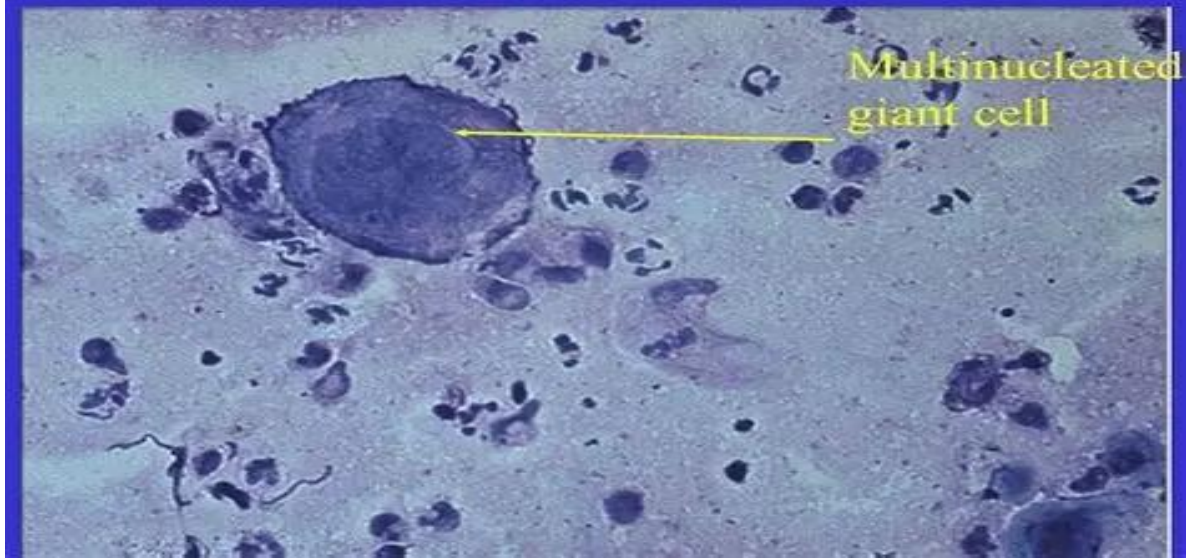
3-Polymerase Chain Reaction: Sensitive method for detecting viral DNA.

4-Enzyme-Linked Immunosorbent Assay: Detects IgM (recent) and IgG (past) antibodies.

5-Direct Fluorescent Antibody Test: Detects viral antigens directly from lesion samples.



Tzanck Smear



Viral Culture Inoculation

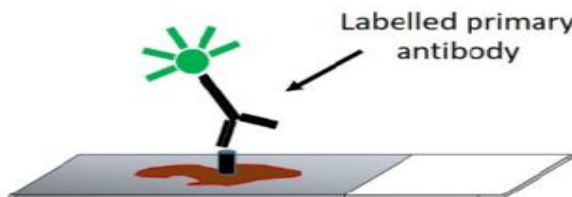
Tissue containing
pathogen antigen

Antigen



Direct fluorescent
antibody test

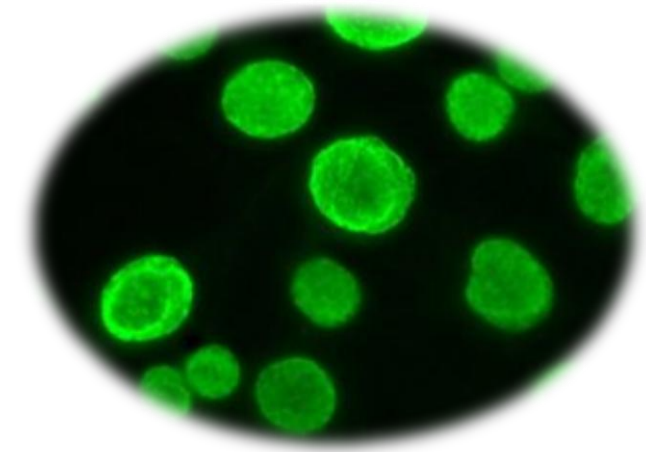
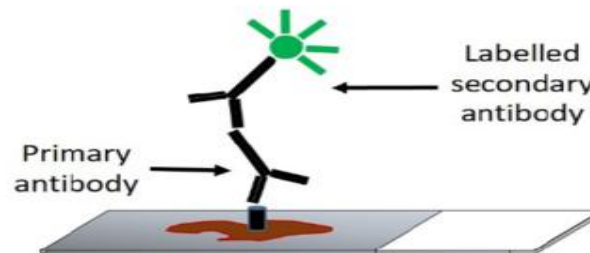
Labelled primary
antibody



Indirect fluorescent
antibody test

Labelled
secondary
antibody

Primary
antibody



HIV (Human Immunodeficiency Virus):

Morphology and Structure

family: Retroviridae

Shape: Spherical with a diameter of approximately (100-120) nm.

Envelope: A **lipid bilayer membrane** with glycoproteins (gp120 and gp41), which act as "spikes" to attach to human host cells.

Capsid: A **bullet-shaped protein shell** (made of protein p24) that protects the virus's genetic material.

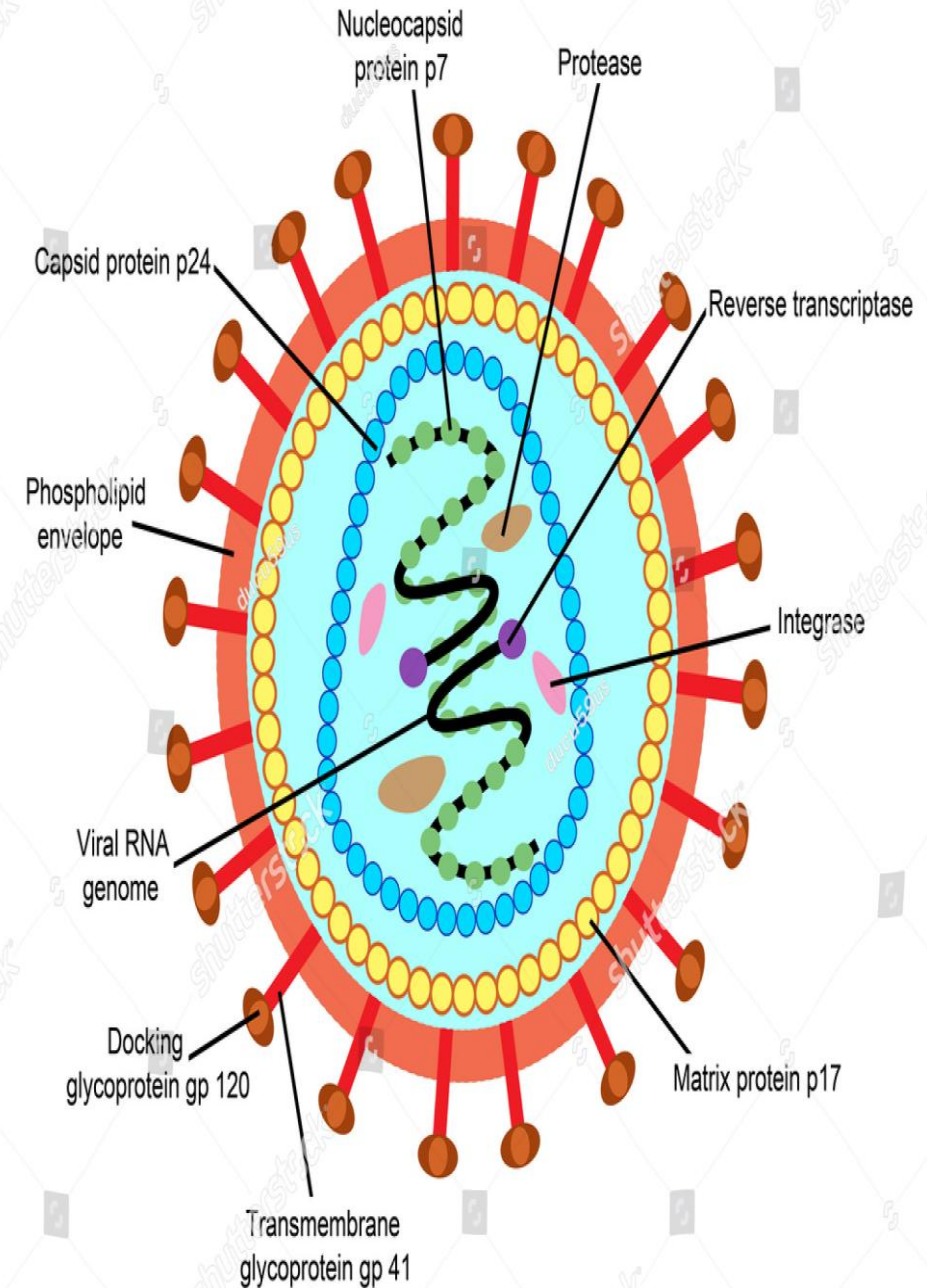
Genetic Material: It contains two copies of single-stranded RNA.

Key Enzymes:

Reverse Transcriptase: Converts viral RNA into DNA.

Integrase: Inserts the viral DNA into the host cell's DNA.

Protease: Cuts long protein chains into smaller, functional proteins to create new viruses.



Virulence Factors:

1-Targeting CD4+ T Cells: By destroying T-Cell, the virus paralyzes the body's ability to fight off other infections.

2-Genetic Integration (Latency): Once the virus integrates its DNA into the human genome, it can remain "**hidden**" for years, making it nearly impossible for the immune system to find the virus.

3-High Mutation Rate: This helping it escape the immune system's antibodies and develop resistance to medications.

4-Massive Replication: the virus produces billions of copies daily, causing a rapid decline in the immune system's defenses.

5-Immune Exhaustion: the chronic inflammation, consume the immune system's resources, leading to AIDS (Acquired Immunodeficiency Syndrome).

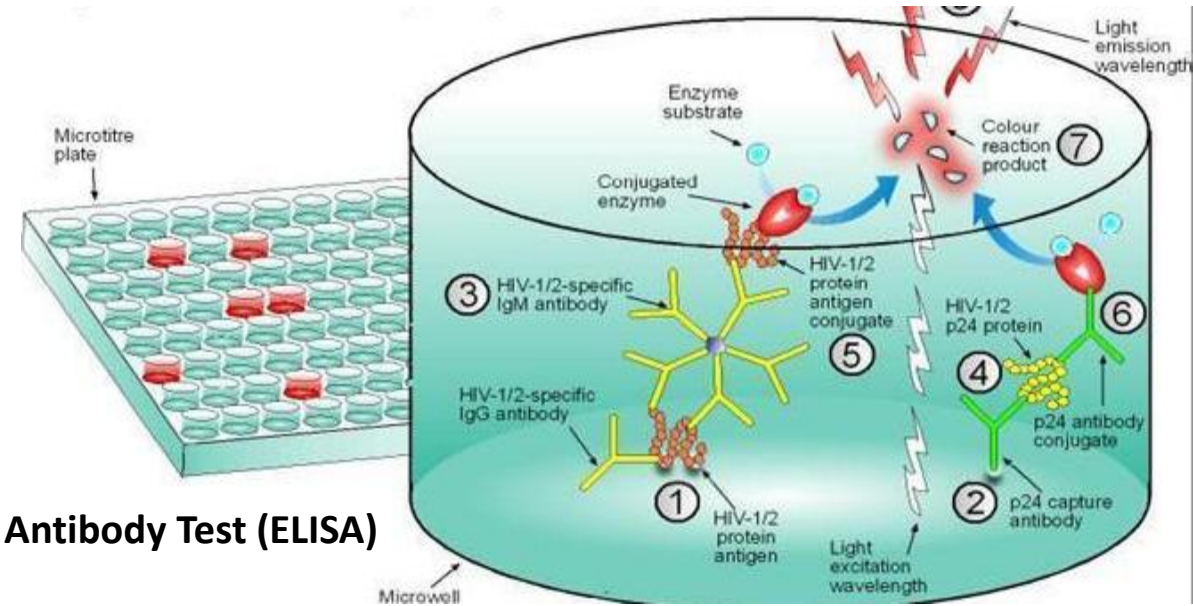
Diagnostic Methods:

1-Antibody Test (ELISA): Screens for antibodies produced by the immune system against HIV.

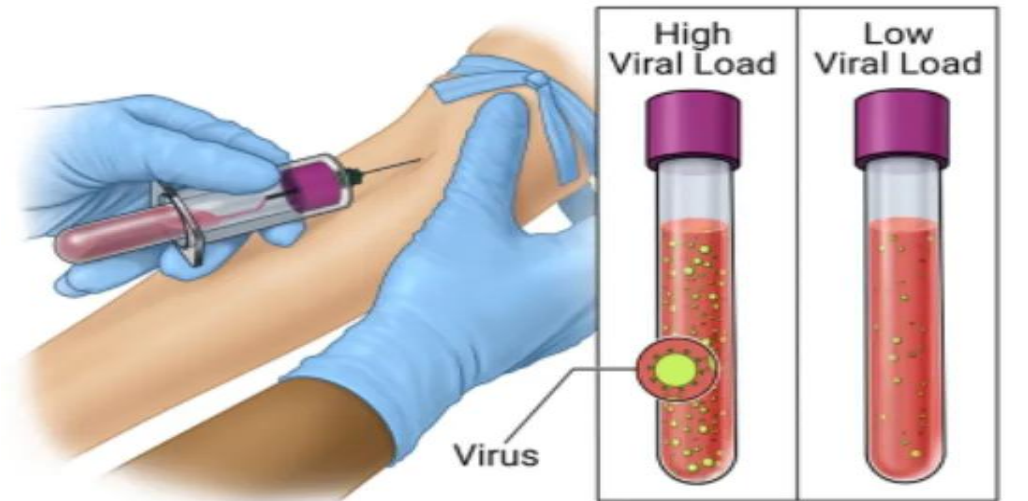
2-Antigen Test (p24): Can detect the virus earlier than the antibody test.

3-Viral Load Test (PCR): Directly measures the amount of viral genetic material in the blood, it is the most accurate.

Antibody Test (ELISA)



Antigen Test (p24)



Viral Load Test (PCR)

Symptoms of Acute HIV



Fever



Chills



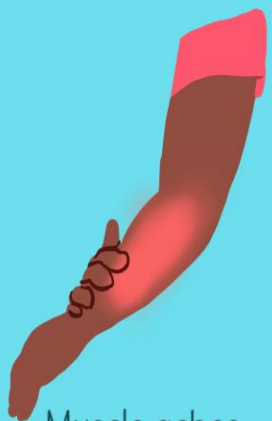
Headaches



Night Sweats



Sore throat



Muscle aches
and pains



Joint pain



Fatigue



Swollen lymph
nodes, mainly
on the neck



Mouth ulcers



اعراض
الايدز
الاولية
بالترتيب

Modes of Transmission:

1-Sexual transmission: The principal route worldwide, occurring through unprotected sexual contact.

2-Blood exposure: Transmission may occur through transfusion of contaminated blood or sharing needles and syringes.

3-Mother-to-child transmission: Infection can be transmitted during pregnancy, delivery, or breastfeeding.

4-Contaminated sharp instruments: Use of non-sterile equipment for tattooing, piercing, or other procedures may facilitate transmission.

Prevention Methods:

1-Safe Practices: Avoiding high-risk behaviors and maintaining monogamous relationships.

2-Personal Hygiene: Never sharing razors, toothbrushes, or needles.

3-Pharmacological Prevention: Using PEP (Pre-Exposure Prophylaxis) for high-risk individuals or PEP (Post-Exposure Prophylaxis) in emergencies.

4-Prevention of Mother-to-Child Transmission (PMTCT): through strict adherence to Antiretroviral Therapy (ART) during pregnancy.



Any Question?



Thank you